MIDTERM ASSIGNMENT

Machine Learning-Based Demand Prediction

For Factory Production Planning of Shampoo, Body Wash, and Laundry Detergent (Regression)

Prepared for: Machine Learning Course

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Presentation Roadmap

Required midterm proposal components covered in a logical defense flow.

1. Topic & Problem

Setting the context for factory production challenges.

2. Objectives & RQs

Defining the scope and research questions.

3. Dataset & Features

Identifying data sources, features, and the target variable.

4. Literature Review

Reviewing forecasting themes and related ML practice.

5. Methodology & Metrics

Explaining algorithms, pipeline, and evaluation criteria.

6. Outputs & Timeline

Detailing tools, execution plan, roles, and risk management.

Goal: show a meaningful, practical, and academically sound ML project proposal.

Introduction: Why This Problem Matters

Daily-use products need accurate production planning to avoid stock problems.

Factory planning challenge: Shampoo, body wash, and laundry detergent demand can change because of season, price, promotion, stock level, and customer orders.

📈Overproduction

Manual planning may produce too much.

Impacts: More stock, higher storage cost, slower cash flow, and possible product aging.

📉Underproduction

Manual planning may produce too little.

Impacts: Out-of-stock risk, missed sales, delayed orders, and lower customer satisfaction.

💡Machine Learning Opportunity

Use historical product and sales data to predict future demand and recommend a better production quantity before the factory produces.

Problem Statement

Converting a real factory planning issue into a regression ML task.

Current Situation

Production planning may depend on experience or simple monthly averages. Demand changes across product types and seasons. Stock and order information may not be fully used in decisions.

Proposed ML Solution

Train models on historical product, production, stock, and sales data. Predict future demand quantity for each product category. Use results to support production planning and inventory decisions.

> **Target Variable:**
> Demand quantity / sales quantity for the next period.

Product Scope: Medtherm Factory Items

The scope stays focused and manageable for a midterm-to-final ML project.

1. Shampoo

Hair cleaning product.
Historical monthly sales, price, promotions.

2. Body Wash

Body washing liquid.
Demand pattern by season and customer orders.

3. Laundry Detergent

Shirt/clothes washing product.
Stock level and production planning needs.

Objectives and Research Questions

Clear objectives make the proposal defensible and measurable.

Project Objectives

- Build a supervised regression model to predict product demand.
- Compare at least three ML algorithms using regression metrics.
- Identify useful features (product type, month, stock, price, promo).
- Create a simple output that supports factory production planning.

Research Questions

- **RQ1:** Can factory product data predict future demand quantity?
- **RQ2:** Which algorithm gives the lowest prediction error?
- **RQ3:** Which features appear most useful for production planning?

> **Expected outcome:** Predicted demand + recommended production quantity

Target Users and Benefits

The project is practical because the prediction supports real factory decisions.

1. Factory Manager

Plan production volume accurately to systematically reduce instances of overstock or shortages.

2. Production Team

Prepare schedule, allocate materials, and determine optimal batch size earlier and more efficiently.

3. Sales / Order Team

Understand expected demand trends and establish clear product priorities for customer fulfillment.

4. Inventory Team

Monitor stock levels proactively and avoid unnecessary storage pressure and holding costs.

Practical value: better planning, less waste, fewer stockouts, and clearer decisions.

Dataset Source and Collection Plan

The data plan must be realistic for midterm approval and final implementation.

Primary Plan

Medtherm historical records: Collect monthly or weekly records for product type, sales quantity, production quantity, stock quantity, price, promotion, and customer order quantity.

Backup Plan

Prepared sample dataset: If real records are incomplete, create a structured sample dataset using the same feature columns, then clearly state the limitation in the paper.

Minimum Data Structure

Field	Example
Date / Month	2025-01
Product Type	Shampoo / Body Wash / Detergent
Previous Sales	1,200 bottles
Stock Quantity	350 bottles
Target Demand	**Sales quantity next month**

Features and Target Variable

The model learns from product, time, sales, stock, and business factors.

Input Feature	Type	Purpose
Product type	Categorical	Separates shampoo, body wash, and detergent patterns
Month / season	Datetime	Captures seasonal demand changes
Previous sales quantity	Numeric	Shows historical demand trend
Stock quantity	Numeric	Shows inventory pressure before production
Price / promotion	Numeric + binary	Captures demand changes from discounts
Customer order quantity	Numeric	Reflects confirmed or expected orders

Target variable: Demand quantity / sales quantity for the next week or month.

Literature Review: Demand Forecasting

Related studies and ML practice connect historical data to future production planning.

Theme 1: Demand depends on history

Many forecasting projects start with previous sales because past demand often contains trend and seasonal patterns.

Theme 2: Business factors matter

Price, promotion, stock quantity, customer orders, holidays, and product type can improve prediction when recorded well.

Theme 3: ML supports comparison

Regression models can be compared with MAE, RMSE, and R^2 to choose a useful model for planning decisions.

Connection to this project: Medtherm product data will be treated as a supervised regression problem.

Literature Review: Algorithms for Prediction

The model choice should balance simplicity, accuracy, and interpretation.

Algorithm	Why useful for this project	Limitation
Linear Regression	Simple baseline; easy to explain relationship between features and demand.	May underfit nonlinear demand patterns.
Decision Tree Regressor	Captures nonlinear rules such as product-month-promotion effects.	Can overfit if tree depth is not controlled.
Random Forest Regressor	Combines many trees; often improves accuracy and stability.	Less simple to explain than a single tree.

Proposal decision: Compare all three models and select the best by validation/test performance.

Proposed Methodology

A reproducible ML workflow from data collection to practical interpretation.

1. Collect data
2. Clean & preprocess
3. Feature engineering
4. Train models
5. Evaluate metrics
6. Interpret & recommend

Leakage prevention

Split data before fitting preprocessing steps; learn imputation, encoding, and scaling only from training data.

Reproducibility

Use a fixed workflow in Python, keep clear code, and save model comparison outputs for the final paper.

Algorithms and Evaluation Metrics

Success is measured by prediction error and usefulness for production planning.

Machine Learning Algorithms

- 1. Linear Regression
- 2. Decision Tree Regressor
- 3. Random Forest Regressor

The best model will be chosen after evaluation, not by assumption.

Evaluation Metrics

- **MAE:** average absolute error
- **RMSE:** stronger penalty for large error
- **R^2:** variance explained by the model

Lower MAE/RMSE and higher R^2 are preferred.

Validation plan

Use train-test split or cross-validation. Compare models using the same data split so results are fair and defensible.

Interpretation focus

- Which product demand is easiest/hardest to predict?
- Which features affect the forecast most?
- How can the factory use prediction to plan quantity?

Programming Language, Framework, and Tools

The implementation uses a practical Python ML toolchain.

1. Python

Main programming language for data analysis and model development.

2. pandas / NumPy

Load, clean, summarize, and transform tabular data efficiently.

3. scikit-learn

Train Linear Regression, Decision Tree, Random Forest, and evaluate metrics.

4. Jupyter / Colab

Run notebook experiments and prepare output screenshots for defense.

Final implementation output: notebook + trained model results + tables/charts for paper and defense.

Expected System Outputs

The final project should show evidence, not only explanation.

1. Demand prediction table

Product-level predicted demand for the next period.

2. Model comparison table

MAE, RMSE, and R^2 for each candidate model.

3. Actual vs. predicted chart

Visual check of forecast performance.

4. Production recommendation

Suggested production quantity by product.

Example Output Format

Values will be filled after data collection and model training.

Product	Predicted Demand	Recommended Production
Shampoo	[model output]	[decision output]
Body Wash	[model output]	[decision output]
Laundry Detergent	[model output]	[decision output]

Work Plan and Timeline

A realistic plan helps show that the proposal can be completed.

Step	Week	Tasks
1	Week 1	Approve topic Assign roles
2	Week 2	Collect data Confirm columns
3	Week 3	Clean data Preprocess features
4	Week 4	Train models Compare metrics
5	Week 5	Interpret results Prepare figures
6	Week 6	Write paper Prepare defense

Note: All members must understand the full project for Q&A, even when responsibilities are divided.

Group Members and Roles

Suggested role division for a maximum four-member group.

Member 1: Project Leader

Coordinate tasks, check timeline, prepare defense flow.

Member 2: Data & Preprocessing

Collect data, clean missing values, encode product type, create features.

Member 3: Model Development

Train algorithms, tune parameters, evaluate metrics, create charts.

Member 4: Paper & Slides

Write introduction, literature review, methodology, results, and references.

Possible Limitations & Risk Management

A strong proposal is honest about constraints and how the group will handle them.

Risk / Limitation	Possible Impact	Management Plan
Small dataset	Model may not generalize well	Use simple models first; explain limitation clearly
Missing values	Training data may be incomplete	Use imputation and document missing data
Unrecorded promos/events	Prediction error may increase	Add promotion/season features when available
Data privacy	Customer/order details may be sensitive	Remove personal names and private details
Overfitting	High training score but weak real performance	Use validation/test split and compare baselines

Risk response: keep the project manageable, transparent, and supported by clear evaluation.

Questions?

References

- SEK SOCHEAT. Machine Learning Course: Midterm and Final Group Assignment Guideline. Norton University, 2026.
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- SEK SOCHEAT. Chapter 04: Model Evaluation and Validation. Machine Learning Course, 2026.
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Thank you for your attention.

Thank you